

L-4. Gully Treatment

 Degraded Soil & Land Improvement	 Medium	 High	 	
Measure Type	Cost	Preferences by Community	Hazards	Implementation Time of Measures

Descriptions

Measures	The degraded and barren hill slopes often experience severe soil erosion when fast flowing water running down from upslope cuts through the clayey soil and make deep channels or gullies. Gully treatment is a crucial aspect of soil and water conservation in these areas.								
	Structural measures, vegetation layers, diversion of water are some techniques of reducing gully erosion and land management. Construction of check dams reduce water velocity and trap sediments resulting in stabilization of gully beds. Digging trenches and terraces along natural contours help collect, divert and slow down the water thus stabilize the sloping land. Additionally, covering gully walls with stones or other erosion preventing materials and planting vetiver grass, napier grass, or other deep-rooted along the contours strengthen the soil and reduce formation of gullies.								
	These gully treatment methods promote soil conservation, reduce soil erosion, enhance water quality, increase agricultural production and assist landscape restoration.								
Location	R-114	R-99	R-65	K-98	K-76	K-71	B-173	B-77	B-76
	■	■	■	■	■	■	■	■	■
Applicable Sites	• Steep, Eroded Hill Slopes • Deforested or Degraded Lands – • Agricultural Lands Affected by Runoff • Riverbanks and Watersheds								
Slope	Less than 3%		3% to 15%		15% to 30%		30% to 60%		Above 60%
	√		√		√				
Design options and performance	• Check Dams (Loose Boulder, Gabion, or Brushwood) – Slows water velocity, traps sediments, and stabilizes gully beds. • Contour Trenches & Terracing – Collects runoff, reduces erosion, and improves water infiltration. • Gully Head Stabilization – Prevents gully deepening using rock bunds, earthen embankments, and vegetative barriers. • Riprap or Stone Pitching – Protects gully walls from further erosion by reinforcing slopes with stones. • Vegetative Cover (Vetiver, Napier Grass, Bamboo) – Strengthens soil structure, controls erosion, and enhances slope stability.								

	Water Diversion Channels – Redirects excess runoff away from vulnerable gully zones.			
Feasibility criteria	- Choosing measures based on the rainfall frequency, intensity and associated runoff velocity which cause the gullies - Considering the topography and slope gradient to select appropriate gully treatment method. For example, steeper slopes may require check dams or contour bunding while flatter slopes need reforestation or grass planting. - Considering the accessibility of the area alongside downslope conditions since construction of check dams or digging trenches can be difficult in remote areas. - Ensuring proper drainage and diversion of water to protect the downslope area from waterlogging			
Operation and maintenance	Routine Land Management – Regular weeding, erosion control, and vegetation monitoring. Soil Fertility Enhancement – Application of organic compost, green manure, and soil conditioners. Seasonal Adjustments – Rainwater harvesting during monsoons, controlled grazing during dry periods. Community-Based Monitoring – Establishing local forest management groups for sustained oversight.			
Cost	Medium			
Benefits				
	Type	Low	Medium	High
	Canopy Cover Increase			
	Carbon Sequestration			
	Crop Production Increase			
	Crop Failure Risk Reduction			
	Ensuring Food Security			
	Fruit Production Increase			
	Forestry Goods Production Increase			
	Livestock Production Increase			
	Fisheries Production Increase			
	Alternative Livelihood Generation			
	Increase of Income			
	Soil Erosion Control			
	Improved Soil Health			
	Soil Fertility Increase			
	Sustainable Water Supply			
	Water Quality Improvement			
	Water Purification			
	River Bank Erosion Control			
	Surface Runoff Reduction			

	Groundwater Recharge			
	Pest and Disease Management			
	Reduction of Pesticides and Fertilizer Use			
	Ecosystem and Biodiversity Increase			
	Better Communications			
	Better Education			
	Better WASH facilities			
Implementation Challenges				
	Type	Low	Medium	High
	Land unavailable/high land area required			
	Supply chain disconnection			
	Land right related problems			
	High human investment capital need			
	Insufficient infrastructures			
	Less demand of agroforestry product			
	Environmentally unsustainable			
	Socially unacceptable			
	Less beneficial			
Environmental performance	- Gully treatment stabilizes degraded land and conserves soils through reducing soil erosion. - Reduction in soil erosion and sedimentation improves the water quality downslope and benefits irrigation, consumptive water use, WASH etc. - Reduction of soil erosion and improvement in water quality enhances land productivity as well as the agroforestry yield. - Gully treatment benefits the ecosystem through restoration of landscape and maintaining ecological balance.			
Remarks	This measure will aid landscape and ecological balance restoration in CHT of Bangladesh.			